Area-weighted historical population

Control: Match of Historical to Modern Geographic Units — Area-weighted population. At State level, this is a historical-instrument choice.

1. Allocate population using geographic overlap. Let P_{jt} be the population of historical reporting area j in census year t, and a_{jt} its mapped area. Most reporting areas are counties. Let $B_{jg,t}$ be the part inside target region g. The allocated historical population is

$$P_{g,t}^{\text{area}} = \sum_j P_{jt} \frac{B_{jg,t}}{a_{jt}}.$$

An MSA is the union of its current member counties. In State analysis, a region is the union of modern counties assigned to one historical FE group in the dataset. The territory year fixes this grouping; the instrument year selects P_{jt} and its source areas. County matching uses the same target membership. The methods differ in population allocation.

2. The assumption is uniform density within each reporting area. If 30% of a historical county's area lies inside an MSA, the method assigns 30% of that county's population to the MSA. It does not know where people lived within that county. For example, 30% of an area with 10,000 residents contributes 3,000 residents. The method therefore handles boundary crossings mechanically, but can misallocate population if settlement was concentrated on one side of the boundary. It uses geographic area rather than modern population weights.

3. The result is a population total. Under Levels, $z_{g,t} = P_{g,t}^{\text{area}}/1{,}000$ is in thousands of people. Natural log uses $\log P_{g,t}^{\text{area}}$ for positive totals. The total instruments modern employment or employment density, according to the selected approach. It is not divided by modern area. It also differs from Weighted Density Overlap, which averages historical county densities. The minimum-overlap threshold used by the overlap instruments does not apply here.

4. Reporting and geographic coverage matter. In State analysis, sum the recorded populations that can be allocated by area. A recorded zero counts as known. An unknown population or a source without a usable polygon contributes no estimate. The resulting sum can be partial. We use it as observed, without scaling it up for the missing area. County and MSA totals retain their source completeness checks.

State estimates use the same geographic screen for all historical instruments. Coverage is the area with a recorded population divided by the full mapped area of the region's member counties. It must meet the chosen minimum in both historical years. For 1790, 12 of the 13 states meet 80%, ten meet 95%, and all 13 have usable population with no minimum. Georgia has coverage of 14.4%. These counts describe the sample available for estimation. With 2010 outcomes and 1790 area-weighted population in levels, the standard CH model for these 12 states yields no IV coefficient within the search range $0 < \theta < 32$.